

AI ARTIFICIAL INTELLIGENCE

AI

What is Artificial Intelligence (AI)

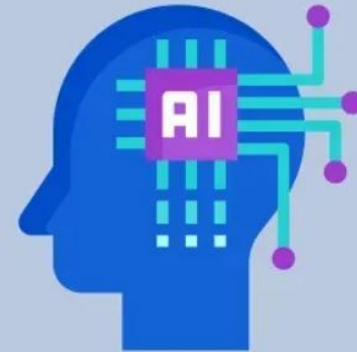
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Artificial Intelligence (AI) is a technology that enables machines and computers to perform tasks that typically require human intelligence. It helps systems learn from data, recognize patterns and make decisions to solve complex problems. It is used in healthcare, finance, e-commerce and transportation offering personalized recommendations and enabling self-driving cars.

What is AI?

- Machines that replicate human-like thinking.
- Analyze vast data to make decisions.



TYPES OF AI



Types of AI

Types of AI Based on Capabilities

- Narrow AI
- General AI
- Super AI

Types of AI Based on Functionalities

- Reactive Machines
- Limit Memory
- Theory of Mind
- Self Aware AI

<https://www.geeksforgeeks.org/artificial-intelligence/types-of-artificial-intelligence/>

USE CASE, APPLICATIONS

4. Autonomous Vehicles

Self-driving vehicles are changing the transportation system and also it plays a important role in the navigation and decision-making processes of these vehicles.

- **Tools:** Tesla Autopilot, Waymo, Uber ATG, Cruise
- **Navigation and Decision-Making:** Vehicles use sensors and algorithms to understand and find roads, detecting objects and obstacles.
- **Safety Improvements:** This technology helps in reducing accidents caused by human error which makes our travel more safer.
- **Traffic Optimization:** Autonomous cars communicate with each other to reduce traffic and improve fuel efficiency.
- **Future of Transport:** These vehicles are more efficient and accessible in the terms of transportation as they are including in delivery and public transport.

5. Healthcare Diagnostics

It is helping doctors to diagnose diseases more quickly and accurately. Lets see how AI tools are used in medical diagnostics.

- **Tools:** Zebra Medical Vision, Aidoc, PathAI, Arterys
- **Medical Imaging Analysis:** They process medical images like X-rays and MRIs to detect diseases early.
- **Personalized Treatment:** Stores patient data and offers personalized treatment plans for our needs.
- **Predicting Health Risks:** Can predict future health risks by analyzing patterns in patient data.
- **Efficiency:** These tools increases the speed of diagnostics and reduces wait times and improving patient outcomes.

7. Finance and Banking

AI plays an important role in making finance and banking services more efficient and secure. Lets see how AI tools are used in this sector.

- **Tools:** Robinhood, Mint, PayPal, Stripe
- **Algorithmic Trading:** These algorithms helps in analyzing market data and make real-time trading decisions.
- **Fraud Detection:** AI monitors financial transactions to detect suspicious activity and prevent fraud.
- **Customer Service:** Also chatbots answer customer questions and helps in assisting with transactions.
- **Personalized Financial Advice:** AI-driven platforms provide investment advices and helps individuals to manage their finances.

10. Gaming and Entertainment

AI is upgrading gaming and entertainment industry by creating more engaging, personalized experiences for users.

- **Tools:** Unity, Unreal Engine, Oculus VR, Steam
- **Dynamic Gameplay:** It can adjust difficulty of games based on player performance also ensures an engaging experience.
- **Content Generation:** It helps in creating game levels, characters and storylines which helps in speeding up game development and offers variations.
- **Smarter NPCs:** It helps in making non-playable characters (NPCs) which behaves like a human and reacts to player actions in a natural way.
- **Immersive Experiences:** In virtual reality (VR) and augmented reality (AR), it adapts environments in real time to provide a better experience.

★ 1. AI in Marketing — (7-Mark Answer)

Introduction:

AI in marketing uses data-driven algorithms to understand customer behaviour, personalize experiences, predict trends, and automate marketing activities. It improves decision-making by analysing large amounts of customer and market data.

How AI Works in Marketing:

AI models analyse customer purchase history, browsing data, and social media behaviour using machine learning, clustering, and predictive analytics. Natural Language Processing (NLP) helps in sentiment analysis and chat-based support.

Applications:

- **Personalized Recommendations:** Platforms like Amazon and Netflix suggest products based on user behaviour.
- **Customer Segmentation:** AI groups customers into segments (age, interests, spending pattern).
- **Targeted Advertising:** AI optimizes ads for the right audience at the right time.
- **Sentiment Analysis:** Identifies customer opinions from reviews and social media.
- **Marketing Automation:** Email campaigns, product suggestions, and notifications.

Advantages:

- Better customer engagement
- Higher conversion rates
- Cost-effective targeted ads
- Real-time insights

Challenges:

- Requires large amounts of customer data
- Privacy and ethical concerns
- Biased model predictions

Conclusion:

AI helps marketers make smarter, faster and personalized decisions, improving both customer satisfaction and business growth.

★ 2. AI in Banking — (7-Mark Answer)

Introduction:

AI in banking enhances security, improves customer service, automates operations, and supports financial decision-making.

Working:

Machine learning models analyse transaction patterns, customer history, and behavioural data. NLP helps chatbots communicate with customers.

Applications:

- **Fraud Detection:** AI identifies unusual patterns in real time.
- **Risk Assessment:** Predicts the probability of loan default.
- **Customer Support:** Chatbots answer queries instantly.
- **KYC Automation:** Verifies identity using image recognition.

Advantages:

- Faster decision-making
- Better security and fraud prevention
- Improved customer service

Challenges:

- Data privacy
- Complex regulatory requirements

Conclusion:

AI strengthens banking operations, offering secure and intelligent financial services.

★ 3. AI in Finance — (7-Mark Answer)

Introduction:

AI supports financial institutions in forecasting, investment decisions, trading, and managing risks.

Working:

Machine learning models identify market patterns, predict trends, and assist in portfolio management. Deep learning detects anomalies and optimizes investment strategies.

Applications:

- **Algorithmic Trading:** High-speed trading using AI predictions.
- **Portfolio Management:** Robo-advisors recommend investments.
- **Risk Management:** Identifies credit, market, and operational risks.
- **Fraud Analytics:** Detects unusual transactions.

Advantages:

- High accuracy in predictions
- Faster trading decisions
- Reduced financial risks

Challenges:

- Market volatility reduces accuracy
- Requires large financial datasets

Conclusion:

AI transforms finance into a more efficient, data-driven industry.

★ 4. AI in Agriculture — (7-Mark Answer)

Introduction:

AI in agriculture improves crop production, soil monitoring, and disease prevention through data-driven techniques.

Working:

AI models analyse images, weather data, soil data, and crop patterns. Drones and sensors collect real-time field data.

Applications:

- **Crop Disease Detection:** Image-based diagnosis of leaf diseases.
- **Precision Farming:** Optimal water, fertilizer and pesticide usage.
- **Yield Prediction:** Forecasts crop output.
- **Automated Irrigation:** Sensors trigger watering automatically.
- **Weed Detection:** AI robots remove unwanted plants.

Advantages:

- Increased crop productivity
- Reduced costs
- Sustainable farming

Challenges:

- Requires technology infrastructure
- High initial cost

Conclusion:

AI enables efficient and sustainable agriculture by using smart, automated methods.

AI in Healthcare (Medical diagnosis)

Introduction (1–2 sentences)

Artificial Intelligence (AI) in healthcare uses algorithms and models to analyse medical data and assist in diagnosis, prognosis and treatment planning. It combines medical knowledge with machine learning and deep learning techniques to improve accuracy and speed of clinical decisions.

How it works (methods & techniques)

AI systems for diagnosis commonly use supervised learning and deep learning. Medical images (X-rays, CT, MRI) are processed by convolutional neural networks (CNNs) to detect patterns and anomalies. Tabular health records and lab tests are used with classification models (logistic regression, random forest, gradient boosting) to predict disease risk. Natural Language Processing (NLP) extracts information from clinical notes. Training requires labeled datasets, feature engineering, model validation and tuning.

Examples & applications (important — give 2–3 concrete examples)

- **Radiology:** CNNs detect pneumonia, tumors or fractures from X-rays and CT scans with high sensitivity.
- **Pathology:** Image analysis identifies cancerous cells in biopsy slides.
- **Predictive analytics:** Models predict risk of diabetes complications or hospital readmission from EHR data.
- **Support tools:** Clinical decision support suggests likely diagnoses and treatment options to doctors.

Advantages (brief, exam-style bullets)

- Faster diagnosis and screening of large patient volumes.
- Improved detection accuracy, especially subtle patterns.
- Reduces clinician workload and human error.
- Enables early intervention and personalized treatment.

Limitations & challenges (brief, required for full marks)

- Requires large, high-quality labeled datasets; risk of biased training data.
- Lack of interpretability (black-box models) can hinder clinical trust.
- Regulatory, privacy and ethical issues (patient data security).
- Needs integration with clinician workflow and validation across populations.

Conclusion (1 line)

AI is a powerful adjunct for medical diagnosis that can improve outcomes when used responsibly alongside clinicians and with proper validation.



★ 6. AI in Gaming — (7-Mark Answer)

Introduction:

AI enhances the gaming experience by creating intelligent, adaptive, and realistic game environments.

Working:

AI uses algorithms like finite state machines, reinforcement learning, and behaviour trees to control NPC actions and game difficulty.

Applications:

- **NPC Behaviour:** Enemies adapt based on player actions.
- **Game Difficulty Adjustment:** Game becomes harder or easier automatically.
- **Procedural Content Generation:** AI generates maps, levels, and challenges.
- **Player Analytics:** Identifies player patterns for game improvements.

Advantages:

- Improved realism and engagement
- Dynamic gameplay
- Reduced manual game design effort

Challenges:

- Requires complex algorithms
- Risk of unpredictable behaviour

Conclusion:

AI makes games more interactive, realistic, and personalised.

★ 7. AI in Space Exploration — (7-Mark Answer)

Introduction:

AI supports space missions by enabling autonomous navigation, data analysis, and robot-assisted exploration.

Working:

AI processes large volumes of space data, controls robots, and assists spacecraft in making decisions without human input.

Applications:

- **Mars Rovers:** NASA's Curiosity uses AI to navigate terrain.
- **Satellite Image Analysis:** Detects climate patterns and planetary features.
- **Autonomous Spacecraft:** AI helps with docking, landing, and navigation.
- **Astronaut Assistance:** AI robots help astronauts during missions.

Advantages:

- Reduced risk to human astronauts
- Faster data processing
- Better mission accuracy

Challenges:

- Technology failures in harsh space conditions
- Limited real-time communication with Earth

Conclusion:

AI makes deep space exploration safer, smarter, and more efficient.

★ 8. AI in Autonomous Vehicles — (7-Mark Answer)

Introduction:

Autonomous vehicles use AI to sense the environment, make decisions, and control driving without human input.

Working:

AI integrates data from cameras, LiDAR, radar, and sensors. Deep learning models identify objects, predict movement, and plan safe driving paths.

Applications:

- Lane detection
- Traffic sign recognition
- Pedestrian detection
- Collision avoidance
- Self-parking systems

Advantages:

- Reduces accidents
- Supports elderly and disabled people
- Less traffic congestion

Challenges:

- Ethical decision-making
- Weather affects sensors
- Safety concerns

Conclusion:

AI-powered autonomous vehicles can transform transportation with safer and smarter mobility.

★ 9. AI in Chatbots — (7-Mark Answer)

Introduction:

AI chatbots interact with users using Natural Language Processing (NLP) and machine learning.

Working:

Chatbots understand user input through NLP, classify intent, and generate suitable responses. Advanced chatbots use deep learning and transformer models.

Applications:

- Customer service
- Online shopping support
- Banking queries
- Medical triage
- Virtual assistants (Alexa, Google Assistant)

Advantages:

- Available 24×7
- Quick responses
- Handles multiple users simultaneously

Challenges:

- Limited understanding of emotions
- May give wrong responses
- Needs constant training

Conclusion:

AI chatbots improve communication efficiency by providing fast, automated, and intelligent support.

IMPORTANCE OF AI

Why does Artificial Intelligence matter?

AI matters for several reasons:

1. **Efficiency and Automation:** Automating repetitive tasks can boost efficiency and productivity, freeing up organizations to concentrate on more intricate and strategic activities.
2. **Insights and Decision Making:** AI can analyze large datasets to uncover patterns, trends, and insights that humans might miss. This can lead to better decision-making and strategic planning.
3. **Personalization:** AI can be used to personalize experiences for customers, such as personalized recommendations in e-commerce or personalized healthcare treatments.
4. **Cost Savings:** By automating tasks and improving efficiency, AI can help organizations save costs in various areas of their operations.
5. **Innovation:** AI is driving innovation in various industries, from healthcare to finance to transportation. It is enabling the development of new products, services, and business models.
6. **Competitive Advantage:** Organizations that leverage AI effectively can gain a competitive advantage by being more agile, innovative, and customer-focused.
7. **Addressing Complex Problems:** AI can be used to address complex problems that are difficult for humans to solve, such as climate change, healthcare challenges, and cybersecurity threats.

<https://www.geeksforgeeks.org/artificial-intelligence/importance-of-artificial-intelligence/>

AI ALGORITHMS



What are AI Algorithms?

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines, allowing them to perform tasks that typically require human cognitive functions such as learning, reasoning, problem-solving, perception, and decision-making.

AI systems analyze vast amounts of data, adapt to new information, and enhance their performance autonomously over time. The ultimate goal of AI is to develop machines that can think, reason, and act autonomously, and in some cases, surpass human capabilities across various domains.

AI algorithms encompass a diverse range of techniques, enabling machines to learn from data, identify patterns, make decisions, and solve complex problems autonomously. Categorized based on functionality and learning approach, these algorithms include:

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- [1. AI Search & Optimization Algorithms](#)
- [2. AI Supervised Learning Algorithms](#)
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- [4. AI Algorithms Based on Neural Networks](#)
- [5. AI Reinforcement Learning Algorithms](#)
- [6. AI Algorithms for Computer Vision](#)
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<https://www.geeksforgeeks.org/artificial-intelligence/ai-algorithms/>

TYPES OF ML

Types of Machine Learning

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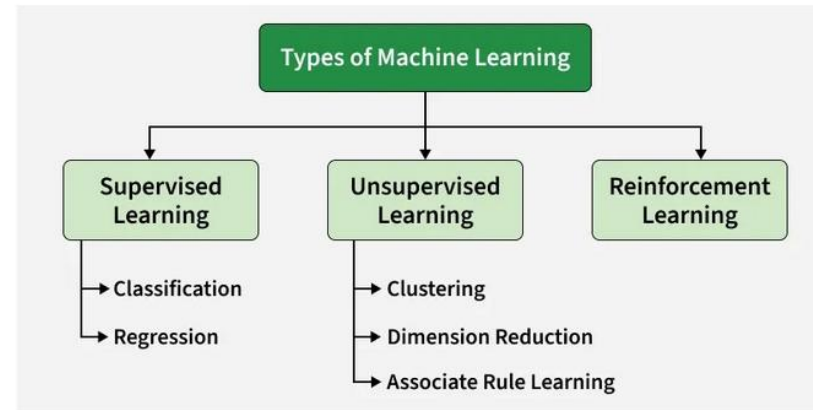


Machine Learning (ML) is a subfield of Artificial Intelligence (AI) that focuses on building algorithms and models that enable computers to learn from data and improve with experience without explicit programming for every task. In simple words, Machine Learning teaches systems to learn patterns and make decisions like humans by analyzing and learning from data.

There are several types of machine learning, each with special characteristics and applications. Some of the main types of machine learning algorithms are as follows:

1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Reinforcement Learning

Additionally, there is a more specific category called Semi-Supervised Learning and Self-Supervised Learning, which combines elements of both supervised and unsupervised learning



Types of Machine Learning

<https://www.geeksforgeeks.org/machine-learning/types-of-machine-learning/>

TYPES OF PROBLEMS SOLVED IN AI

Problem Solving in Artificial Intelligence

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Problem solving is a fundamental concept in [artificial intelligence \(AI\)](#), where systems are designed to identify challenges, make decisions and find efficient solutions. AI uses agents which are systems that perceive their environment and take actions to achieve specific goals. They go beyond simple reflex agents which react to stimuli based on pre-defined rules. Instead, problem-solving agents actively analyze situations, evaluate different options and choose the best action to reach their goal.

These agents work by:

- **Perceiving the environment:** They collect data about their surroundings such as sensor inputs or observations.
- **Defining the problem:** They clearly understand the problem including the starting point, the available actions and the desired goal.
- **Exploring different possibilities:** They consider various ways to solve the problem and evaluate which approach is likely to succeed.
- **Evaluating and deciding:** Once they explore options, they assess the outcomes and pick the best course of action based on factors like time, resources and success likelihood.
- **Learning and adapting:** Many problem-solving agents can learn from past experiences, improving their decision-making abilities over time.

Types of Problems in AI

AI systems can face different types of problems. These problems are categorized based on their impact on the system's performance and how they can be handled:

<https://www.geeksforgeeks.org/artificial-intelligence/problem-solving-in-artificial-intelligence/>

ADV AND DISADV OF AI

Advantages and Disadvantage of Artificial Intelligence

- Advantages of Artificial Intelligence
 - 1. Efficiency and Productivity
 - 2. Enhanced Decision Making
 - 3. Personalization and Customer Experience
 - 4. Innovation and Development
 - 5. Risk Reduction
- Disadvantages of Artificial Intelligence
 - 1. Job Displacement
 - 2. High Implementation Costs
 - 3. Ethical and Privacy Concerns
 - 4. Dependency and Reliability
 - 5. Lack of Emotional Intelligence
- Conclusion

<https://www.geeksforgeeks.org/artificial-intelligence/advantages-and-disadvantage-of-artificial-intelligence/>

Artificial Creativity

AI creativity refers to the ability of artificial intelligence to generate novel concepts, ideas, and works, often by combining familiar concepts in new ways or exploring conceptual spaces. While AI can produce creative output, it currently lacks human traits like consciousness, emotions, and cultural context, and its creativity is fundamentally dependent on human input and the data it was trained on. Therefore, many view AI as a powerful tool that enhances human creativity rather than a true replacement for it, but its role and the definition of creativity are evolving. 

How AI demonstrates creativity

- **Generating new ideas:** AI can analyze vast datasets to find unique combinations of concepts, which can then inspire human artists and writers.
- **Visual art:** AI can be used to generate original images, patterns, and compositions that artists can use as a starting point or incorporate into their work.
- **Music composition:** AI can compose melodies, harmonies, and entire pieces of music by analyzing existing compositions and creating new musical ideas.
- **Writing:** AI can generate text prompts, ideas, or even full paragraphs to help writers and poets overcome writer's block.
- **Interactive experiences:** Artists can create interactive art where the audience can use prompts to guide the AI in generating content. 

Limitations of AI creativity

- **Lack of consciousness and emotion:** AI does not have personal desires, dreams, or emotions; it can mimic them but does not genuinely feel them, which is a core component of human creativity.
- **Dependence on data:** All AI-generated content is based on existing human-created data, leading some to argue it is "parasitic" on human creativity rather than truly original.
- **Lack of cultural understanding:** AI struggles with nuances like satire, subtext, and cultural context that are deeply embedded in human expression.
- **Need for human input:** AI requires human prompts and direction to function. Its outputs are a result of its programming and the data it was fed, and it cannot independently decide to become an artist. 

The future of AI and human creativity

- **A collaborative tool:** AI is increasingly seen as a tool that enhances human creativity, streamlining tasks and providing inspiration, rather than a replacement.
- **New creative processes:** The rise of tools like text-to-image models is introducing new workflows, even if artists desire more control than simple text prompts.
- **Ethical considerations:** Concerns exist about the ethics of using AI, particularly regarding the data used for training and how it impacts artists' intellectual property.
- **Focus on the human element:** Despite AI's advancements, the human element—authenticity, personal expression, and individual connection—remains crucial to the value and meaning of art. 

AI Tools & Frameworks

★ AI Tools and Frameworks — Detailed 7-Mark Answer

Artificial Intelligence (AI) development relies heavily on specialized tools and frameworks that simplify model creation, training, and deployment. These tools provide optimized mathematical operations, allow efficient use of hardware accelerators (GPU/TPU), and include built-in algorithms for machine learning (ML), deep learning (DL), and natural language processing (NLP). Below are the most widely used and influential AI tools, explained in detail.

● 1. TensorFlow

TensorFlow is an open-source deep learning framework developed by Google Brain in 2015.

Key Features

- **Computation Graphs:** Uses static computation graphs, enabling optimized execution and deployment.
- **Cross-Platform Support:** Works on CPU, GPU, and Google TPUs for extremely fast training.
- **High Performance:** Efficient for large-scale deep learning models.
- **TensorBoard:** A visualization toolkit for monitoring loss curves, accuracy graphs, histograms, and model graphs.
- **TF Lite & TF Serving:**
 - TF Lite → deploy models on mobile/IoT devices
 - TF Serving → deploy AI models in production servers

Common Use Cases

- Image classification (CNNs)
- NLP models (RNNs, Transformers)
- Object detection (SSD, YOLO variants)
- Speech recognition
- Large-scale production AI systems

2. PyTorch

PyTorch is a deep learning framework developed by Facebook AI Research (FAIR).

Key Features

- **Dynamic Computation Graph:**
 - Graph is created on the fly, making debugging easier.
 - Very useful in research environments.
- **Pythonic API:** Feels like writing normal Python code.
- **Autograd Engine:** Automatically computes gradients for backpropagation.
- **Strong Research Community:** Most recent research papers use PyTorch.
- **TorchScript:** Allows converting PyTorch models for optimized deployment.

Use Cases

- Transformer-based models (BERT, GPT)
- GANs (image generation)
- Reinforcement learning
- Robotics and control systems
- Academic research & experiments

3. Scikit-learn

A lightweight but powerful ML library for classical machine learning algorithms.

Key Features

- Implements almost all standard ML algorithms:
 - Classification → SVM, Logistic Regression, Naive Bayes
 - Regression → Linear Regression, Ridge, Lasso
 - Clustering → K-Means, DBSCAN
 - Dimensionality reduction → PCA
- Built-in tools for:
 - Data preprocessing
 - Train-test splitting
 - Feature scaling
 - Model evaluation (accuracy, precision, recall, ROC)
- Very easy for beginners and fast on small-medium datasets.

Use Cases

- Credit scoring
- Customer segmentation
- Spam classification
- Medical data analysis
- Predictive analytics

4. Keras

Keras is a **high-level neural network API** that runs on top of TensorFlow.

Key Features

- **User-friendly:** Extremely simple code for building deep learning models.
- **Modular:** Clear structure of layers, models, optimizers, and losses.
- **Prototyping:** Ideal for quickly building and testing models.
- Supports CNNs, RNNs, Autoencoders, GANs, Transformers.

Use Cases

- Image recognition
- Sentiment analysis
- Time-series forecasting
- Beginners learning deep learning

5. OpenCV (Open Source Computer Vision Library)

A widely used computer vision library with C++, Python, and Java support.

Key Features

- Thousands of functions for image and video processing:
 - Filtering, edge detection, thresholding
 - Face and object detection (Haar cascades)
 - Optical flow
 - Real-time tracking
- Integrates with deep learning frameworks for advanced applications.
- Highly optimized for real-time applications.

Use Cases

- Face recognition systems
- Surveillance
- Gesture detection
- Autonomous vehicle vision pipeline
- Medical image preprocessing



● 6. SpaCy

A modern, production-ready NLP library.

Key Features

- **High-speed tokenization** (much faster than NLTK)
- Pre-trained pipelines for:
 - POS tagging
 - Dependency parsing
 - Named Entity Recognition (NER)
- Built-in word vectors
- Supports training custom NLP models
- Excellent for large-scale industry applications

Use Cases

- Text classification
- Chatbots
- Information extraction
- Document processing in enterprises

7. NLTK (Natural Language Toolkit)

An academic and research-oriented NLP library.

Key Features

- Huge collection of corpora (WordNet, movie reviews, Brown corpus)
- Traditional NLP tools:
 - Tokenizers
 - Stemmers (Porter, Lancaster)
 - Lemmatizers
 - Taggers and chunkers
- Good for teaching and experimenting with NLP algorithms.

Use Cases

- Learning NLP fundamentals
- Building rule-based NLP systems
- Educational chatbots
- Linguistic analysis

● 8. Gensim

A powerful library for topic modeling and vector space modeling.

Key Features

- Implements:
 - Word2Vec
 - Doc2Vec
 - FastText embeddings
 - LDA (Latent Dirichlet Allocation) topic modeling
- Efficient for handling large text corpora.
- Streaming and incremental training support.

Use Cases

- Search engines
- Semantic similarity detection
- Document clustering
- Topic extraction in news, social media

● 9. HuggingFace Transformers

A state-of-the-art NLP framework for transformer-based models.

Key Features

- Provides pretrained models like BERT, GPT-2, RoBERTa, T5.
- Supports fine-tuning on custom tasks.
- Integrates with PyTorch and TensorFlow.
- Easy pipeline API for quick inference.

Use Cases

- Text classification
- Machine translation
- Text summarization
- Chatbot development
- Question answering systems



Artificial Intelligence vs Machine Learning vs Deep Learning

Difference Between Artificial Intelligence vs Machine Learning vs Deep Learning

Last Updated : 07 Aug, 2024



Artificial Intelligence is basically the mechanism to incorporate human intelligence into machines through a set of rules(algorithm). AI is a combination of two words: "Artificial" meaning something made by humans or non-natural things and "Intelligence" meaning the ability to understand or think accordingly. Another definition could be that "**AI is basically the study of training your machine(computers) to mimic a human brain and its thinking capabilities**".

AI focuses on 3 major aspects(skills): **learning, reasoning, and self-correction** to obtain the maximum efficiency possible.

Machine Learning:

Machine Learning is basically the study/process which provides the system(computer) to learn automatically on its own through experiences it had and improve accordingly without being explicitly programmed. **ML is an application or subset of AI**. ML focuses on the development of programs so that it can access data to use it for itself. The entire process makes observations on data to identify the possible patterns being formed and make better future decisions as per the examples provided to them. **The major aim of ML is to allow the systems to learn by themselves through experience without any kind of human intervention or assistance.**

Deep Learning:

Deep Learning is basically a sub-part of the broader family of Machine Learning which makes use of **Neural Networks**(similar to the neurons working in our brain) to mimic human brain-like behavior. DL algorithms focus on **information processing patterns** mechanism to possibly identify the patterns just like our human brain does and classifies the information accordingly. DL works on larger sets of data when compared to ML and the **prediction mechanism is self-administered by machines.**

<https://www.geeksforgeeks.org/artificial-intelligence/difference-between-artificial-intelligence-vs-machine-learning-vs-deep-learning/>